Lab Experiment: 5

Title: Association Analysis and Rule Generation Using Apriori Algorithm

Course title: ***Data Mining Problem Description***

Course code: CSE-452
4^{th} Year 1^{st} Semester Examination 2025

Date of Submission: 05/03/2026

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1. Objective

The objective of this experiment is to learn how to perform **Association Rule Mining** using the **Apriori Algorithm in WEKA**.

The experiment helps to discover relationships or dependencies between items in a dataset such as **Bread with Butter, Diaper with Beer, Bat with Ball**, etc.

Association analysis is widely used in **market basket analysis** to understand customer purchasing behavior.

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2. Tools and Software Used

- WEKA Data Mining Tool
- Microsoft Excel
- Dataset: Customer Purchase Dataset

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3. Dataset Description

A **Customer Purchase Dataset** is used where each row represents a transaction and each column represents a product item.

The dataset contains the following items:

- Bread
- Butter
- Milk
- Banana
- Diaper
- Beer
- T-Shirt
- Trouser
- Bat
- Ball

These items are chosen because they have natural dependencies such as:

- Bread → Butter
- Diaper → Beer
- T-Shirt → Trouser
- Bat → Ball

4. Dataset Sample

TID	Bread	Butter	Milk	Banana	Diaper	Beer	TShirt	Trouser	Bat	Ball
1	Yes	Yes	Yes	No	No	No	No	No	No	No
2	Yes	No	Yes	Yes	No	No	No	No	No	No
3	No	No	No	No	Yes	Yes	No	No	No	No
4	Yes	Yes	No	No	No	No	Yes	Yes	No	No
5	No	No	No	No	No	No	No	No	Yes	Yes
6	Yes	Yes	Yes	No	No	No	No	No	No	No
7	No	No	No	No	Yes	Yes	No	No	No	No
8	Yes	No	Yes	Yes	No	No	No	No	No	No
9	Yes	Yes	No	No	No	No	Yes	Yes	No	No
10	No	No	No	No	No	No	No	No	Yes	Yes

Total Transactions (N) = **10**

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5. Steps of the Experiment

Step 1

Prepare the dataset in **Microsoft Excel** with product items and transaction records.

Step 2

Save the dataset in **CSV format**.

Example:

`customer_purchase.csv`

Step 3

Open the CSV file and convert it into **ARFF format** for WEKA.

Example:

`customer_purchase.arff`

Step 4

Open **WEKA Explorer** and load the ARFF dataset.

Step 5

Go to the **Associate tab**.

Step 6

Select the algorithm:

```
Apriori
```

Step 7

Click **Start** to run the Apriori algorithm.

Step 8

WEKA generates **Association Rules** with support and confidence values.

Step 9

If the generated rules are not efficient, adjust parameters such as:

- Minimum Support
- Minimum Confidence

Then run the algorithm again.

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6. Apriori Algorithm

Apriori is a **frequent itemset mining algorithm** used to discover **association rules between items**.

It works in two main steps:

1. Find frequent itemsets using **minimum support**.
2. Generate association rules using **minimum confidence**.

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7. Support and Confidence

Support

Support measures how frequently an itemset appears in the dataset.

$\text{Support}(X \cup Y) = \sigma\{(X \cup Y)/N\} * 100$

Where:

- $\sigma(X \cup Y)$ = number of transactions containing both X and Y
- N = total number of transactions

Confidence

Confidence measures how often rule **X → Y** is correct.

$\text{Confidence}(X \rightarrow Y) = \{\sigma(X \cup Y)/ \sigma X\} * 100$

Where:

- $\sigma(X)$ = number of transactions containing X

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8. Generated Association Rules

Some rules generated from the dataset are:

Rule 1

Bread → Butter

Support calculation:

Bread & Butter appears in **3 transactions**

Support = (3/10)*100= 30\%

Confidence calculation:

Bread appears in **5 transactions**

Confidence = (3/5)* 100 = 60\%

Rule 2

Diaper → Beer

Diaper & Beer appears in **2 transactions**

Support = (2/10)*100 = 20\%

Diaper appears in **2 transactions**

Confidence = (2/2) * 100 = 100\%

Rule 3

Bat → Ball

Bat & Ball appears in **2 transactions**

Support = (2/10)* 100 = 20\%

Bat appears in **2 transactions**

Confidence = (2/2)* 100\%

Rule 4

T-Shirt → Trouser

Support = **20%**

Confidence = **100%**

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9. Result

The Apriori algorithm successfully discovered relationships between items in the dataset.

Strong association rules such as **Diaper → Beer**, **Bat → Ball**, and **T-Shirt → Trouser** were generated with high confidence values.

These rules show the dependency between products in customer transactions.

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10. Conclusion

In this experiment, association rule mining was performed using the **Apriori algorithm in WEKA**.

The algorithm identified meaningful relationships between items in the dataset.

Association analysis is very useful in **market basket analysis, recommendation systems, and business decision making**.

Thus, the Apriori algorithm effectively helps in discovering hidden patterns in large datasets.